👩‍🏫 Staff Guide · Horticulture 1 · Unit 6 Seeds (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
Her readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as her worksheets for that lesson
- Quick check: score, then scale to her test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Seeds Quiz**: Paper quiz not in the export; our quiz is team-written.
- **Seeds Worksheet / Special Treatment of Seeds / Seed Stories**: Title only in the export.
- **Month-Long Seed Experiment**: Skipped for Horticulture 1 (saved for Horticulture 2).

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

🌰 Seeds: Guided Notes

Horticulture 1 · Unit 6 · Level 3 (closer to the class work) · goes with Unit 6 Definitions, Parts of a Seed, Seeds Worksheet/Quiz, Seed Tri-Fold

Name: _____

Use the class slides and your reading. Key words:

propagation | *seed coat* | *embryo* | *endosperm* | *germination* | *radicle* | *cotyledon* | *true leaves* | *scarification* | *stratification* | *direct seeding* | *indirect seeding*

A. Seed parts

1. Draw and label a bean seed: seed coat, embryo, cotyledons.

2. The _____ stores food for the embryo.

B. Germination

3. Seeds need water, warmth and _____.

4. Monocots have _____ cotyledon; dicots have _____.

5. Explain scarification and why it is used.

_____

_____

C. Seeding and math

6. **Germination rate**

> Rate = seeds sprouted ÷ seeds planted × 100
> Example: 15 of 20 → 75%

a) 9 of 10 sprouted _____

b) 12 of 24 sprouted _____

🌰 Seeds: Reading

Horticulture 1 · Unit 6 · Level 3 (closer to the class work) · goes with Unit 6 Definitions, Parts of a Seed, Seeds Worksheet/Quiz, Seed Tri-Fold

Name: _____

From Seed to Seedling

Propagation is making new plants. The most common way is from seed. A seed has three main parts: the seed coat, a hard covering that protects it; the embryo, the tiny plant that will grow; and the endosperm, stored food that feeds the embryo as it sprouts. In some plants, such as beans, the food is stored in the cotyledons.

Germination begins when a seed takes in water. Seeds also need warmth and air, and some tiny seeds, like petunias, need light. The root (radicle) grows down first, then the shoot pushes up. The first leaves to appear are cotyledons, or seed leaves; monocots have one and dicots have two. The leaves that come next, which look like the adult plant's leaves, are called true leaves.

Some seeds need special treatment before they will germinate. Seeds with very hard coats can be scarified, or nicked and sanded, so water can get in. Others are soaked, or chilled for weeks in a process called cold stratification, which imitates winter.

Growers use two seeding methods. Direct seeding means sowing seeds where the plant will grow, which works well for beans, carrots and corn. Indirect seeding means starting seeds in trays in a greenhouse and transplanting them later, which gives plants like tomatoes and petunias an early start. Seeds should be planted about two to three times as deep as they are wide.

Check your reading

1. Name the three main parts of a seed and the job of each.

2. What is the difference between cotyledons and true leaves?

3. When would a grower choose indirect seeding?

🌱 Sowing Seeds & Transplanting: Lab Sheet

Horticulture 1 · Unit 6 · Level 3 · goes with Garden Seed Propagation (GH participation) + transplanting

Name: _____

Question: How does sowing depth affect germination?

Materials

cell pack, seed-starting mix, bean or marigold seeds, labels, ruler

⚠️ Safety

- Gloves; pre-wet mix; mask if dusty
- Never eat seeds
- Wash hands

Prediction

Which depth will sprout best: shallow, correct, or very deep?

_____

_____

Procedure

1. Sow 3 cells at the correct depth, 3 cells very deep, 3 on the surface.
2. Label each.
3. Water equally; keep warm.
4. Count sprouts each day for 7-10 days.

Data

Depth	Seeds planted	Sprouted	Rate %
Surface	3		
Correct	3		
Very deep	3		

Analysis (Claim, Evidence, Reasoning)

1. Claim: which depth worked best? Evidence (rates) and reasoning (stored food, moisture, light).

🌰 Seeds: Quick Check

Horticulture 1 · Unit 6 · Level 3 (closer to the class work) · goes with Unit 6 Definitions, Parts of a Seed, Seeds Worksheet/Quiz, Seed Tri-Fold

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The part that becomes the new plant is the…

A. Seed coat | B. Endosperm | C. Embryo

2. Dicots have…

A. Two cotyledons | B. One cotyledon | C. No cotyledons

3. Cold stratification imitates…

A. Summer | B. Fire | C. Winter

4. Starting tomatoes in trays and transplanting is…

A. Indirect seeding | B. Direct seeding | C. Grafting

5. 16 of 20 seeds sprouted. What is the germination rate?

_____

6. Why can a seed planted too deep fail to come up?

_____

_____

🌰 Seeds: Guided Notes · ANSWER KEY

Horticulture 1 · Unit 6 · Level 3 (closer to the class work) · goes with Unit 6 Definitions, Parts of a Seed, Seeds Worksheet/Quiz, Seed Tri-Fold

Name: _____

Use the class slides and your reading. Key words:

propagation · *seed coat* · *embryo* · *endosperm* · *germination* · *radicle* · *cotyledon* · *true leaves* · *scarification* · *stratification* · *direct seeding* · *indirect seeding*

A. Seed parts

1. Draw and label a bean seed: seed coat, embryo, cotyledons.
Diagram with seed coat outside, two cotyledons, embryo between.
2. The **endosperm** stores food for the embryo.

B. Germination

3. Seeds need water, warmth and **air** .
4. Monocots have **1** cotyledon; dicots have **2** .
5. Explain scarification and why it is used.
Nicking or sanding a hard seed coat so water can enter and germination can begin.

C. Seeding and math

6. **Germination rate**

> Rate = seeds sprouted ÷ seeds planted × 100
> Example: 15 of 20 → 75%

a) 9 of 10 sprouted **90%**
b) 12 of 24 sprouted **50%**

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🌱 Sowing Seeds & Transplanting: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 6 · Level 3 · goes with Garden Seed Propagation (GH participation) + transplanting

Name: _____

Question: How does sowing depth affect germination?

Materials

cell pack, seed-starting mix, bean or marigold seeds, labels, ruler

⚠️ Safety

- Gloves; pre-wet mix; mask if dusty
- Never eat seeds
- Wash hands

Prediction

Which depth will sprout best: shallow, correct, or very deep?
Correct depth (2-3x width).

Procedure

1. Sow 3 cells at the correct depth, 3 cells very deep, 3 on the surface.
2. Label each.
3. Water equally; keep warm.
4. Count sprouts each day for 7-10 days.

Data

Depth	Seeds planted	Sprouted	Rate %
Surface	3	**1**	**33%**
Correct	3	**3**	**100%**
Very deep	3	**1**	**33%**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: which depth worked best? Evidence (rates) and reasoning (stored food, moisture, light).
Correct depth; deep seeds ran out of food, surface seeds dried out.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🌰 Seeds: Quick Check · ANSWER KEY

Horticulture 1 · Unit 6 · Level 3 (closer to the class work) · goes with Unit 6 Definitions, Parts of a Seed, Seeds Worksheet/Quiz, Seed Tri-Fold

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The part that becomes the new plant is the…

A. Seed coat | B. Endosperm | **C. Embryo**

2. Dicots have…

A. Two cotyledons | B. One cotyledon | C. No cotyledons

3. Cold stratification imitates…

A. Summer | B. Fire | **C. Winter**

4. Starting tomatoes in trays and transplanting is…

A. Indirect seeding | B. Direct seeding | C. Grafting

5. 16 of 20 seeds sprouted. What is the germination rate?

80%

6. Why can a seed planted too deep fail to come up?

It uses up its stored food before reaching light.

🔑 *Team-written (the paper Seeds Quiz is not in the export), please check.*